



Department of Computer Science and Engineering (CSE)

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Course Code : EEE 2124

Course Name : Electronics Laboratory

Experiment No: 06

Experiment Name: Analysis of Diode Clipper Circuits and Transfer Characteristics using DC Sweep

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Submitted By

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Introduction:

Clipper circuits are fundamental electronic networks that limit the amplitude of an input signal to a specific voltage level. This experiment analyzes the input-output behavior, or transfer characteristics (V_{out} vs. V_{in}), of four distinct diode-based circuits—including both unbiased and biased configurations—using the DC Sweep Analysis function in the Proteus simulation software. The objective is to determine the clipping thresholds from the simulation graphs and validate them against theoretical predictions based on the practical diode model ($V_f \approx 0.7 \text{ V}$).

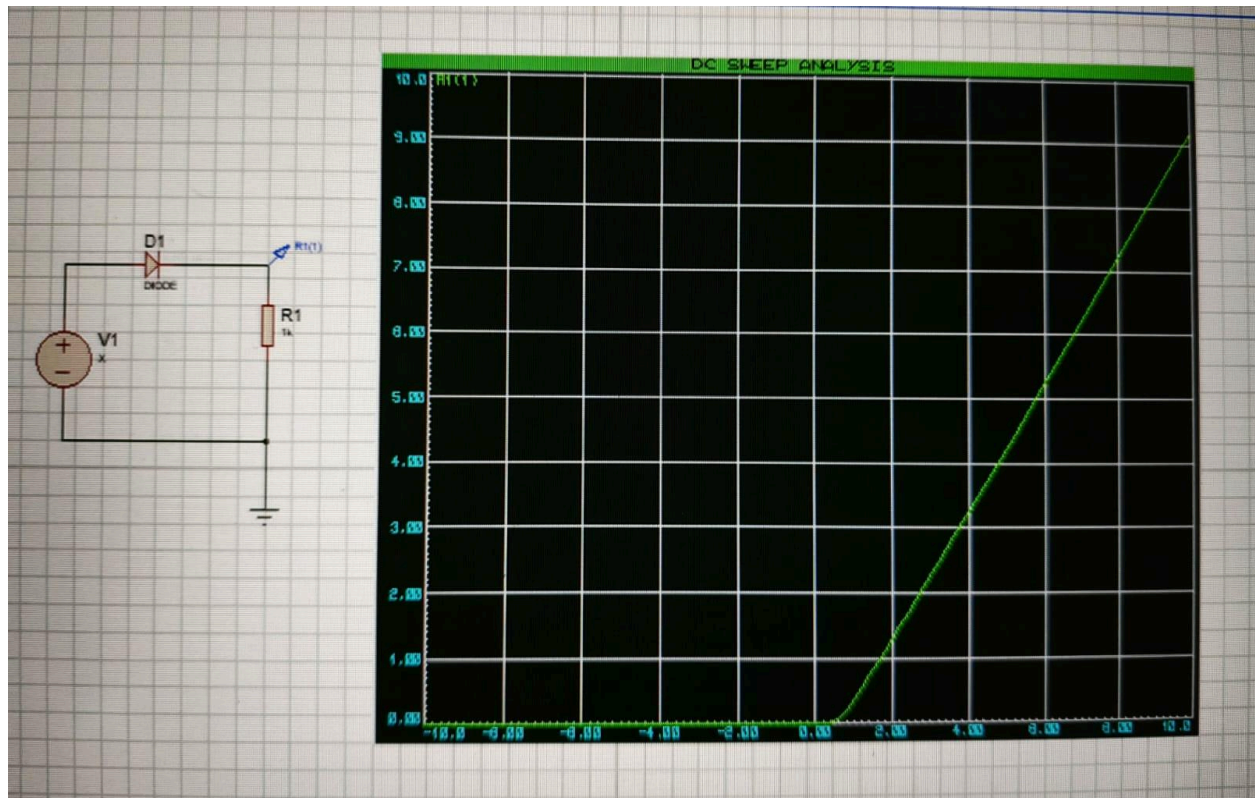
Apparatus:

- Proteus Design Suite
- Laptop/Computer

Results and Analysis:

The analysis is based on the four DC Sweep graphs provided, where the output variable is plotted against the sweeping input voltage.

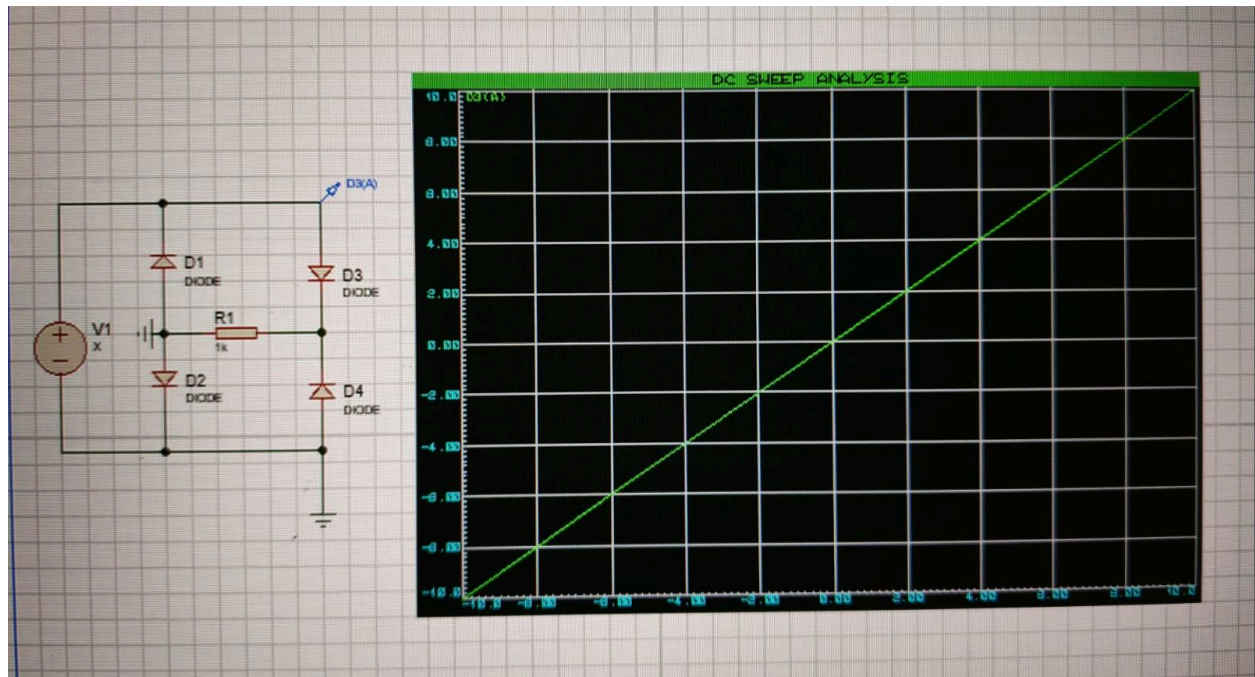
Circuit 1:



Input Condition (V_{in})	Diode State	Output Condition (V_{out})
$V_{in} \leq 0.7$ V	OFF	$V_{out} \approx V_{in}$ (Pass-through)
$V_{in} > 0.7$ V	ON	$V_{out} \approx 0.7$ V (Clamped)

Analysis: The graph's slope abruptly flattens when V_{in} exceeds ≈ 0.7 V. This confirms the diode, acting in shunt, clamps the positive input peaks at its forward voltage drop.

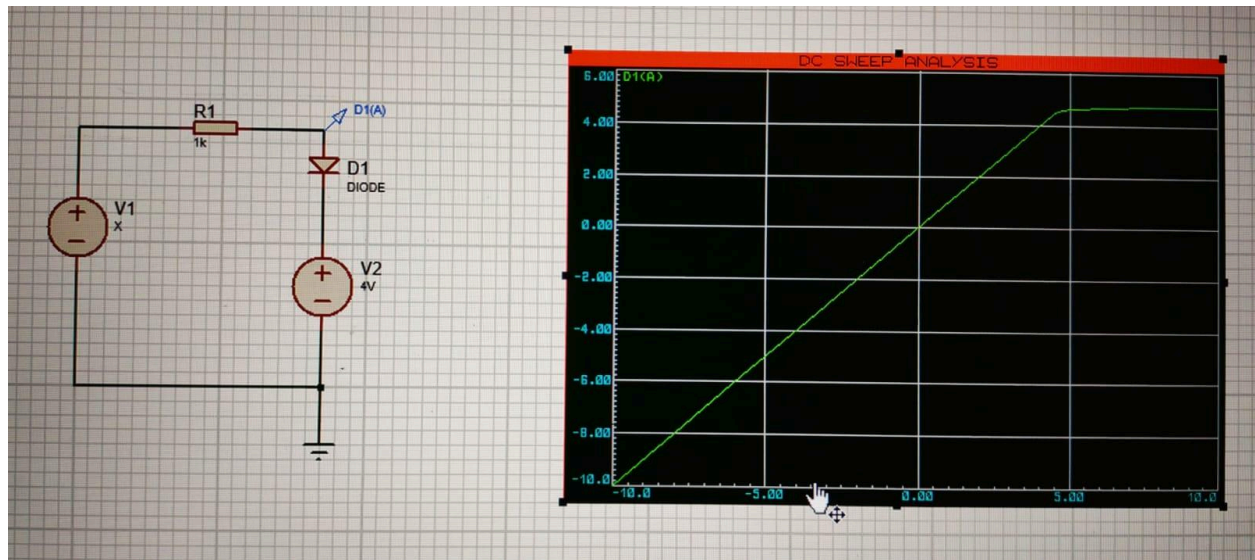
Circuit 2:



Input Condition (V_{in})	Diode State	Output Condition (V_{out})
$V_{in} > 1.4$ V	Two Diodes ON	$V_{out} \approx V_{in} - 1.4$ V (Rectified)
$V_{in} \leq 1.4$ V	All Diodes OFF	$V_{out} \approx 0$ V (Dead Zone)

Analysis: The graph shows a characteristic V-shape with a "dead zone" extending to ± 1.4 V. The output only becomes active when $|V_{in}|$ overcomes the combined forward voltage drop of the two series diodes required for current flow.

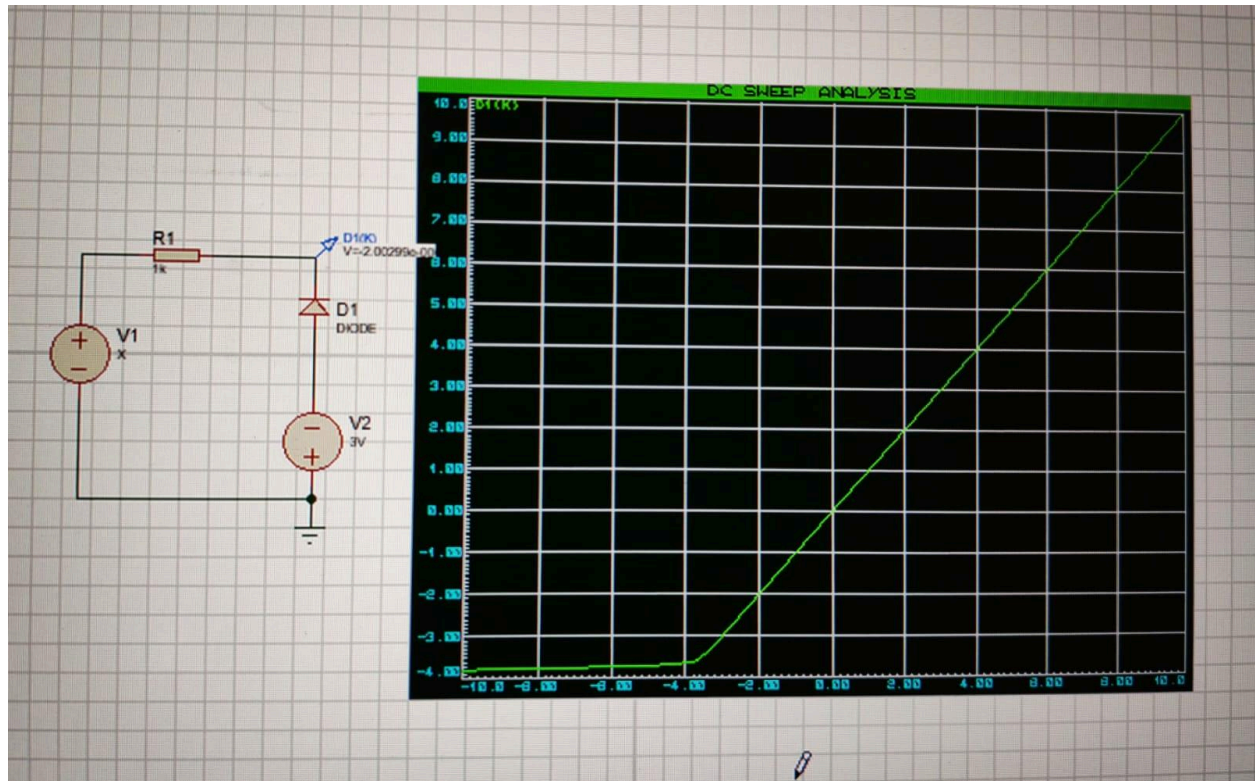
Circuit 3:



Input Condition(V_{in})	Diode State	Output Condition (V_{out})
$V_{in} \leq 4.7$ V	OFF	$V_{out} \approx V_{in}$ (Pass-through)
$V_{in} > 4.7$ V	ON	$V_{out} \approx 4.7$ V (Clamped)

Analysis: The graph is linear until V_{in} reaches ≈ 4.7 V, above which the curve becomes perfectly horizontal. This validates that the maximum output voltage is limited by the precise threshold set by the bias voltage plus the diode drop.

Circuit 4:



Input Condition (V_{in})	Diode State	Output Condition (V_{out})
$V_{in} < -3.7$ V	ON	$V_{out} \approx V_{in} - 3.7$ V (Pass-through)
$V_{in} \geq -3.7$ V	OFF	$V_{out} \approx -3.7$ V (Clamped)

Analysis: The graph shows V_{out} is clamped at a negative voltage near -3.7 V for V_{in} values greater than that threshold. This confirms the lower clipping boundary is precisely determined by the negative sum of the 3 V bias and the 0.7 V diode drop.

Discussion:

The experiment successfully utilized the Proteus DC Sweep Analysis to generate the voltage transfer characteristics, providing empirical confirmation of the theoretical behavior of the diode clipper circuits.

- Unbiased and Biased Clippers: The results confirmed that $V_f(0.7\text{ V})$ dictates the threshold in the unbiased case (Circuit 1). For the biased clippers (Circuits 3 and 4), the clipping levels were accurately shifted to 4.7 V and -3.7 V , validating the fundamental formula $V_{clip}=V_{bias}\pm V_f$.
- Wave-Shaping and Rectification: The Full-Wave Bridge (Circuit 2) illustrated complex wave-shaping, showing a 1.4 V dead zone before the rectification (absolute value function) began.

The consistency between the simulated transfer characteristics and the calculated clipping thresholds validates the design and analysis principles of diode wave-shaping networks, proving the diode's role as a voltage-controlled switch.